

A New Paradigm in Drug Discovery

Multi-target Drugs (Small Molecules) for pain and CNS diseases

August 2026

Vivozon Group Organization



KOSDAQ LISTED COMPANY **OTC / ETC**
Formulation / Production / Sales



First-in-class & Best-in-class
Drug Discovery & Development

Vivozon Canada

Toronto,
J&J Innovation center

Global BD Center

Vivozon US

New Jersey

Global Clinical Center

Medical Writing / RA

- FDA IND / NDA
- FDA Corres.



Vivozon is a biopharmaceutical company in pursuit of breakthrough innovation
for **pain and CNS therapeutics** with **multi-target drug discovery platform**

Pipeline

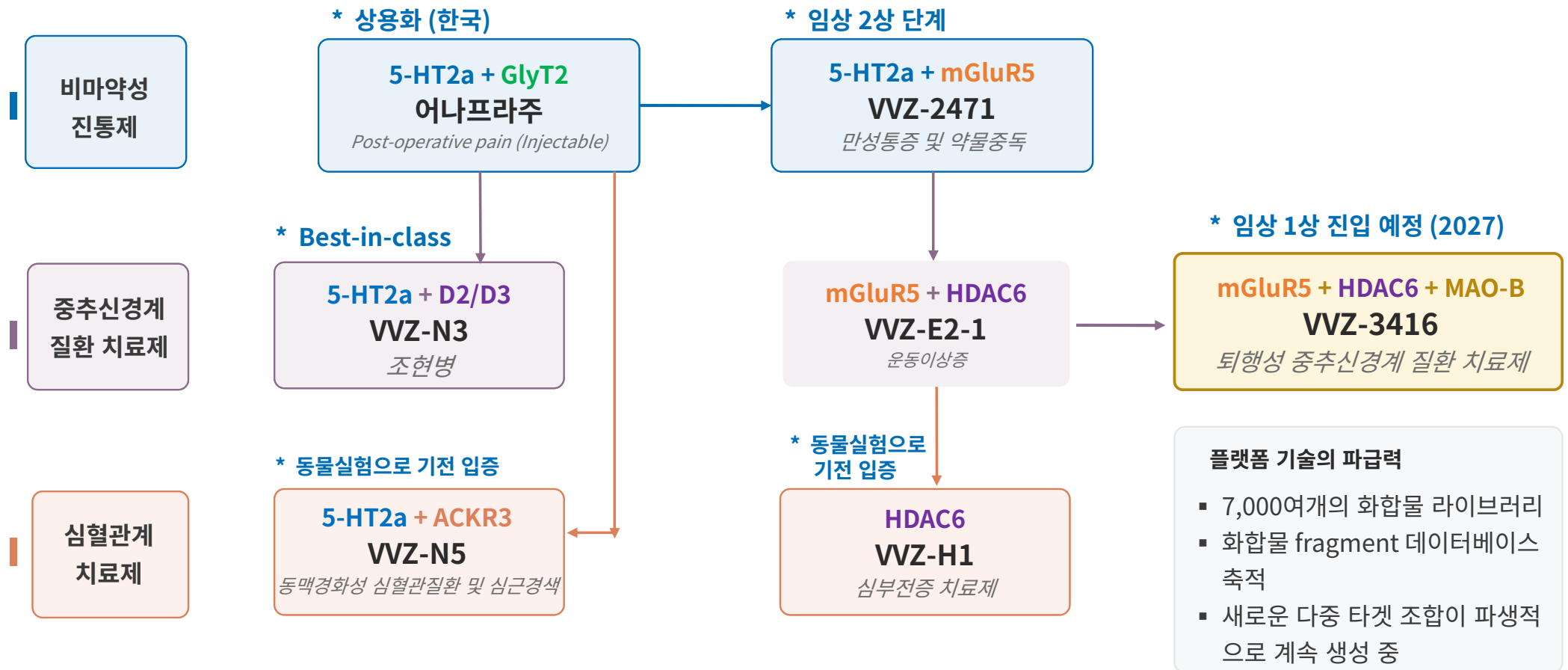


구 분		적응증	진행 단계					향후 계획
			발굴	전임상	1상	2상	3상	
VVZ-149 (오피란제린)	주사제*	수술 후 통증						2024 국내 허가 2026 미국 3상
	외용제*	급성 통증						
VVZ-2471	경구제	만성 통증, 약물 중독						PHN 임상 2a 완료
VVZ-3416	경구제	신경퇴행성 질환 (AD/PD)						2027 임상 1상 예정
VVZ-N3	경구제	조현병						Best-in-class 포지셔닝 확보 (L/O추진)
VVZ-N5	경구제	항혈전						Lead Optimization
VVZ-H1	경구제	심방세동/암						Lead Optimization

* 비보존제약이 국내 전용실시권 보유

Pipeline Based on Platform Technology

“생물학적, 화학적으로 새로운 다중 타겟 치료제의 파생적 도출”



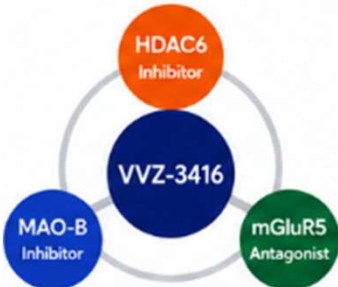
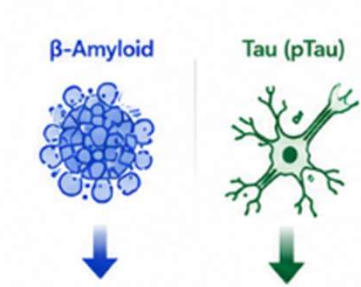
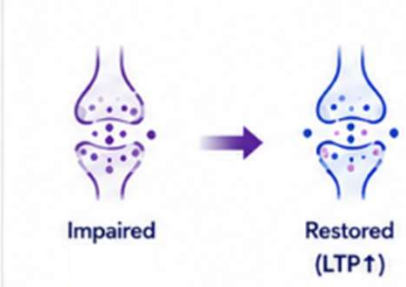
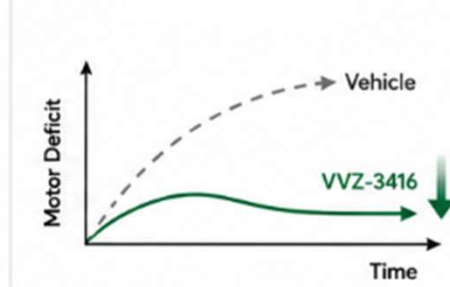
VVZ-3416

NEXT-GENERATION MULTI-TARGET DRUG

- **First-in-Class**
- **Alzheimer's & Parkinson's Disease**
- **Oral Formulation**
- **Current Status: GLP-Tox**

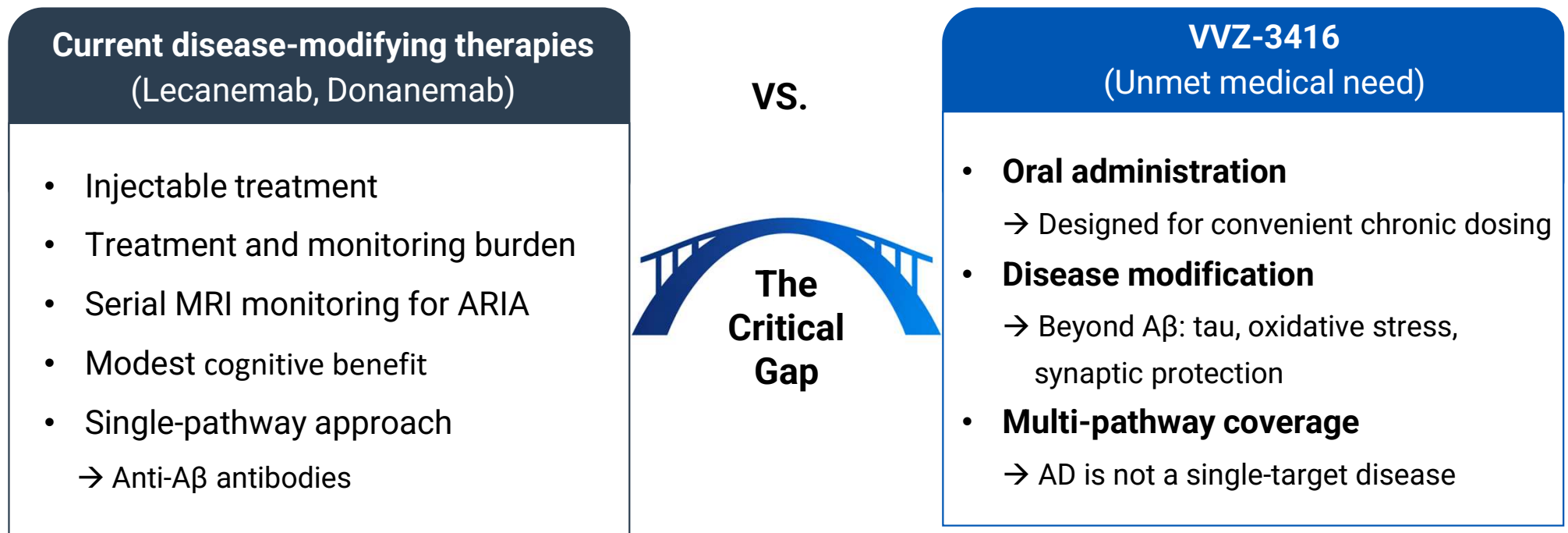
Executive Summary



1	2	3	4	5
Triple-Target Disease Modification	Dual Reduction of Amyloid and Tau	Restoration of Synaptic Function	Disease Progression Control in Parkinson's Disease (6-OHDA Model)	Clinical-Stage Readiness
				- Preclinical Development - GLP Toxicology (Completion in 2026) - First-in-Human (Planned in 2027)
Oral Small Molecule	Improved cognition (Morris Water Maze)	Neuronal Survival Neurite Outgrowth LTP Restoration	- Attenuation of motor decline - Potential to slow disease progression	

Alzheimer's Disease Treatment Landscape

Bridging the critical gap for oral, chronic disease-modifying therapy



VVZ-3416 is designed to address this gap:

An oral, multi-target, CNS small molecule for chronic disease modification

SECTION

01

Mode of Action

Target Potency & Selectivity



Balanced multi-target activity across complementary CNS pathways

mGluR5

HDAC6

MAO-B

mGluR5

IC₅₀ **0.559 μM**

Negative allosteric modulator

Functional Ca²⁺ assay

mGluR 1/2/3/4/6/7/8 : IC₅₀ >10 μM

HDAC6

IC₅₀ **0.104 μM**

HDAC6 selective inhibitor

HDAC6 enzyme inhibition assay

HDAC 1/2/3/4/5/7/8/9 : IC₅₀ >30 μM

MAO-B

IC₅₀ **0.37 μM**

Reversible and selective inhibitor

MAO-B enzyme assay

MAO-A IC₅₀: 47.23 μM · 128-fold selectivity

➤ Off-target liability screening

Kinase panel

Reaction Biology Corp
single-dose 10 μM · 371 kinases

0 / 371

>50% inhibition

GPCR panel

Eurofins DiscoverX gpcrMAX
single-dose 10 μM · 336 assays

1 / 336

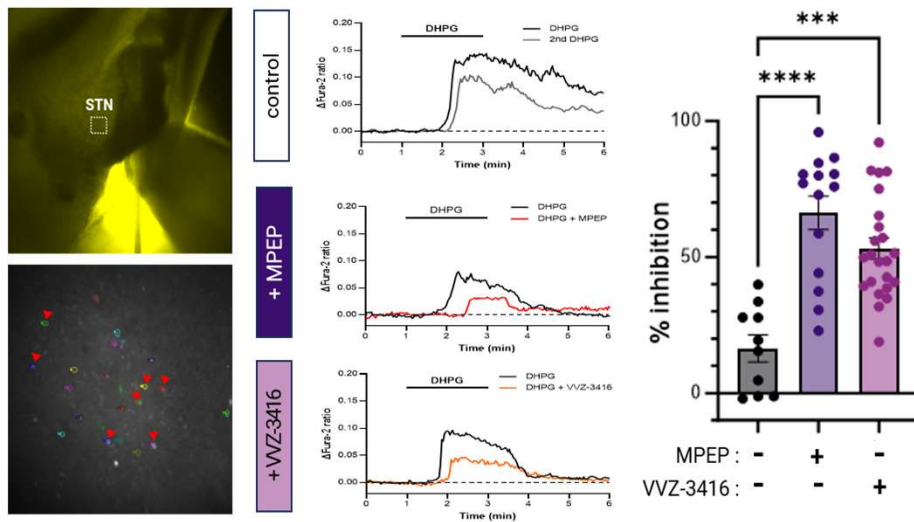
CXCR4: 67% inhibition @ 10 μM
follow-up planned

Functional and Electrophysiological Validation of mGluR5 Antagonism



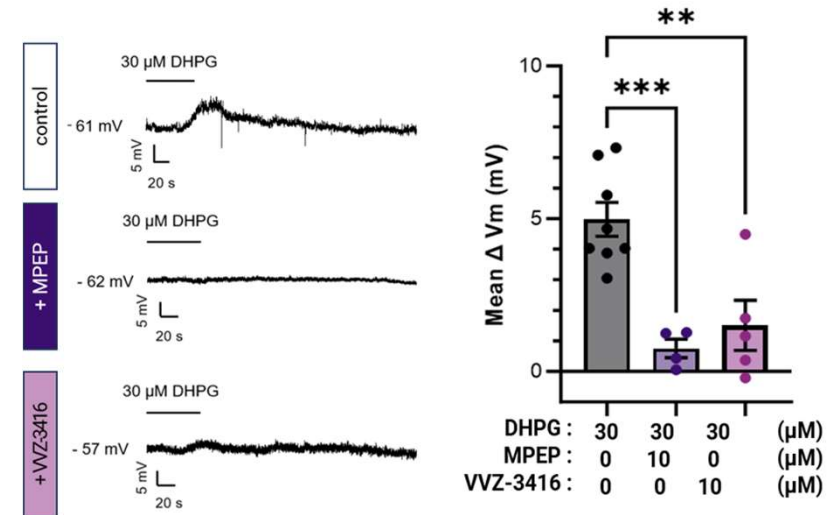
VVZ-3416 suppresses mGluR-5-mediated Ca²⁺ signaling and neuronal depolarization in native STN neurons

➤ Ca²⁺ Imaging



Sample: C57BL/6N (STN), Method: Fura-2AM Ca²⁺ imaging (n=10~23 per group), Drugs: DHPG (mGluR5 agonist), MPEP (mGluR5 NAM)

➤ Whole-Cell Patch Clamp



Sample: C57BL/6N (STN), Method: Whole-cell patch clamp recording (n=4~8 per group), Drugs: DHPG (mGluR5 agonist), MPEP (mGluR5 NAM)

mGluR5 antagonism by VVZ-3416 was confirmed at both Ca²⁺ signaling and neuronal excitability levels

SECTION

02

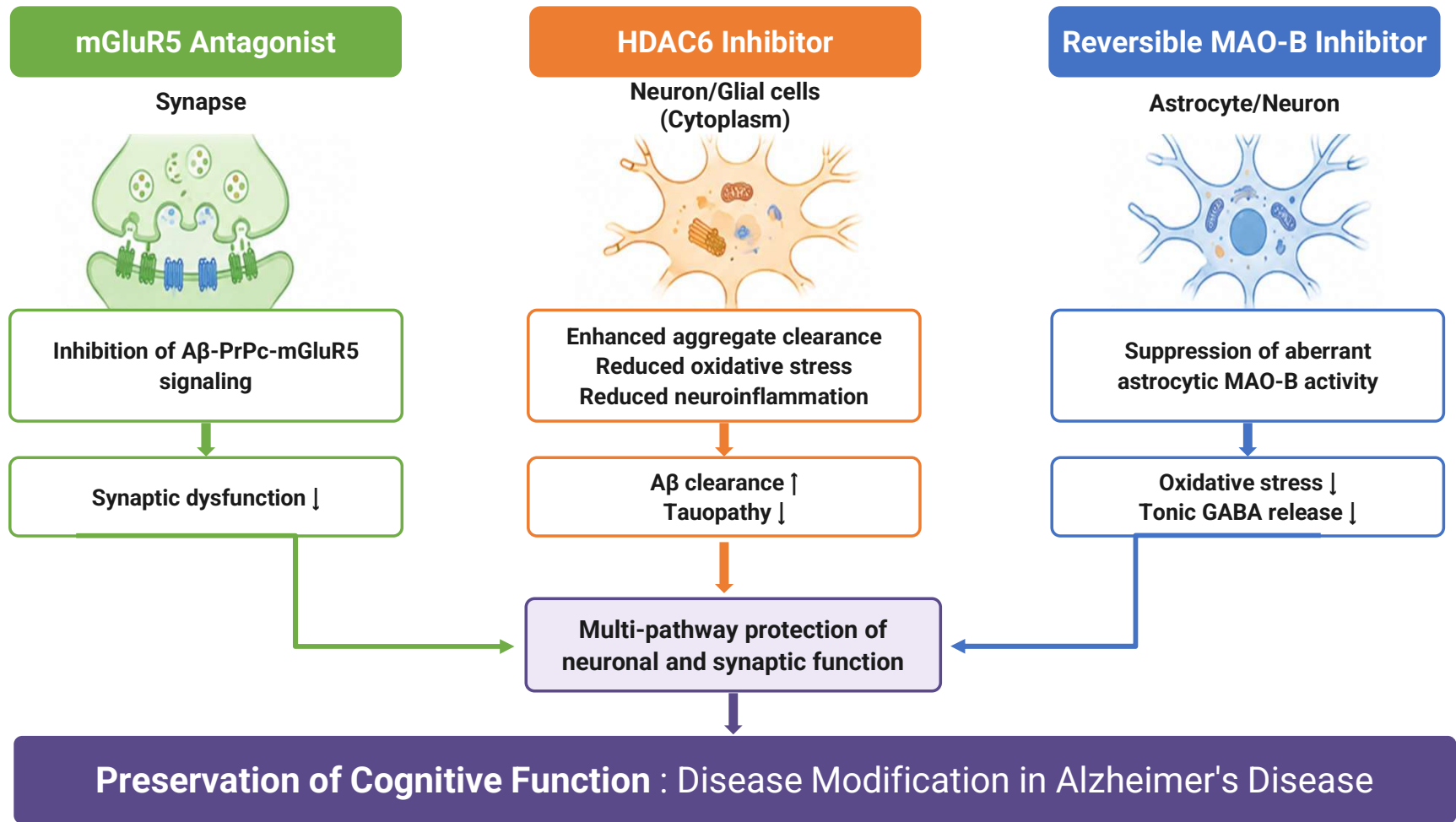
Preclinical Efficacy : Alzheimer's Disease

Multi-Target Disease Modification Strategy for Alzheimer's Disease



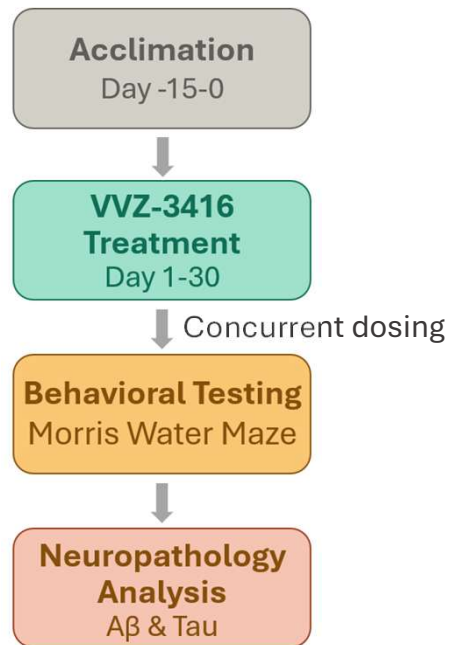
VVZ-3416

**Triple-Target
Small
Molecule**

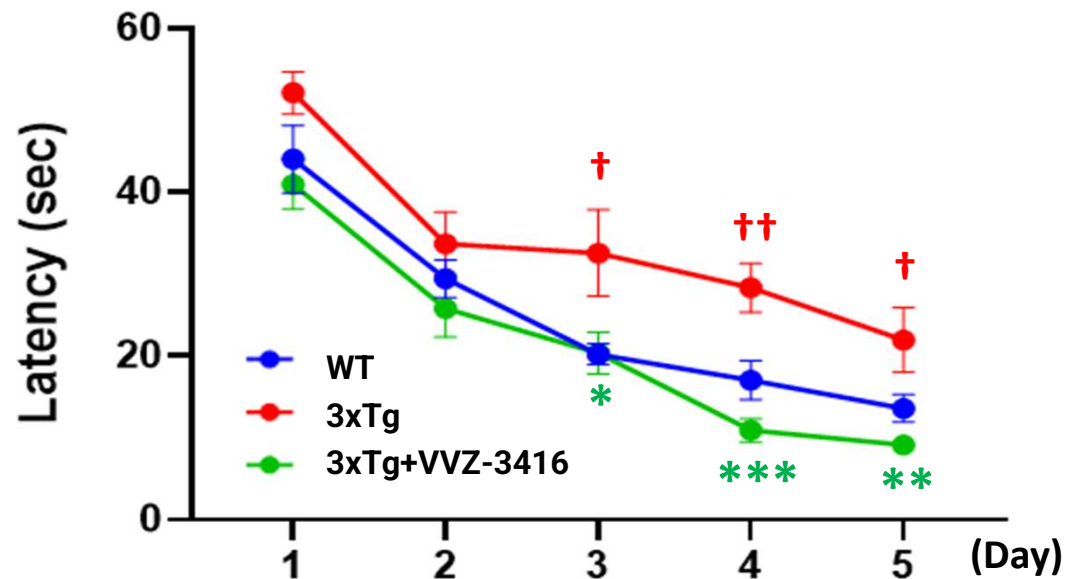


VVZ-3416 Improved Cognitive Function in 3xTg-AD Mice

❖ Procedure

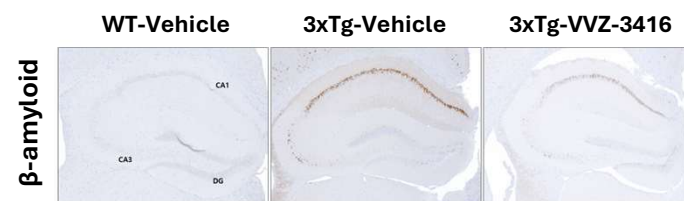


❖ Morris Water Maze Test

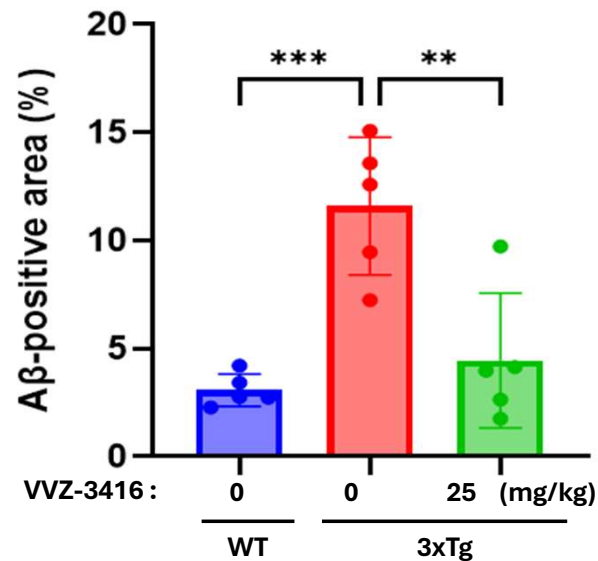


Behavioral findings suggest cognitive benefits of VVZ-3416

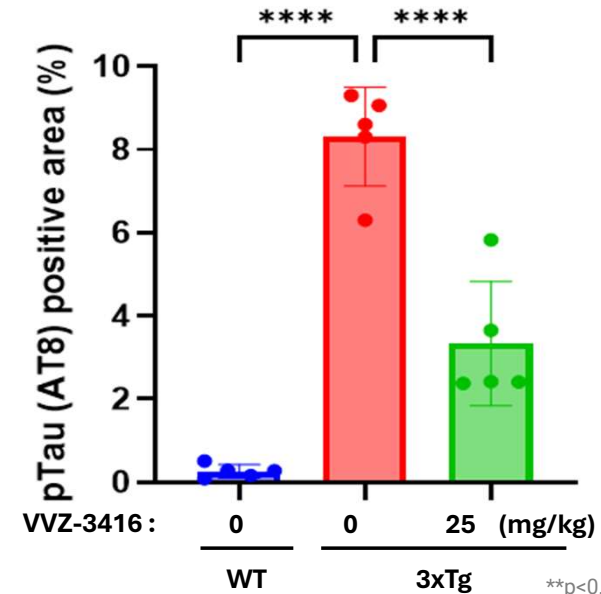
VVZ-3416 Reduced Both Amyloid and Tau Pathology in 3xTg-AD Mice



- Hippocampus: CA1



- Hippocampus: CA1



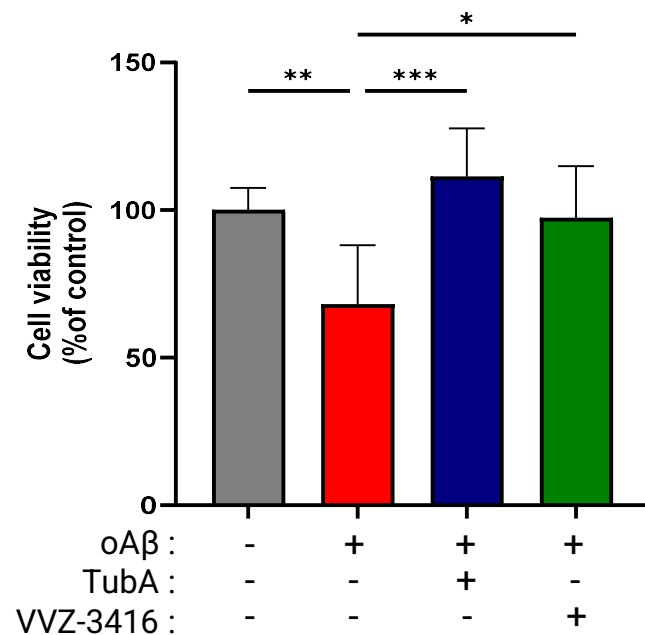
p<0.01, *p<0.001, ****p<0.0001

Reduced amyloid and tau pathology supports the disease-modifying potential of VVZ-3416

VVZ-3416 Supports Neuronal Survival and Neurite Outgrowth

HDAC6

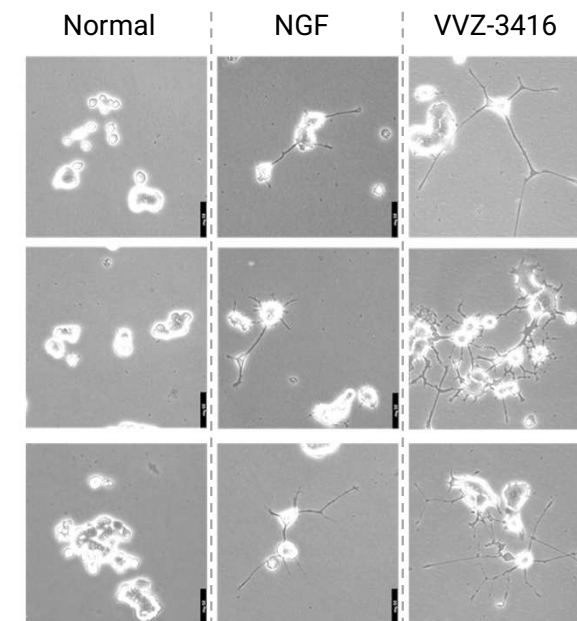
❖ Protection against oA β -induced cytotoxicity (SH-SY5Y)



oA β : oligomeric amyloid beta, TubA: tubastatin A (HDAC6 inhibitor)

*p<0.05, **p<0.01, ***p<0.001, n=6/group

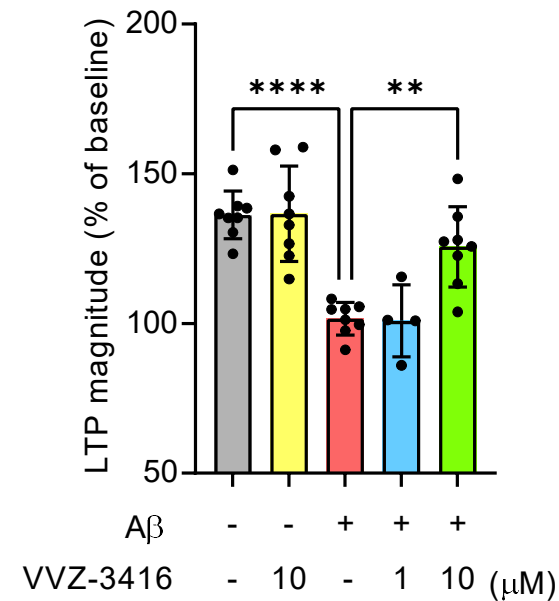
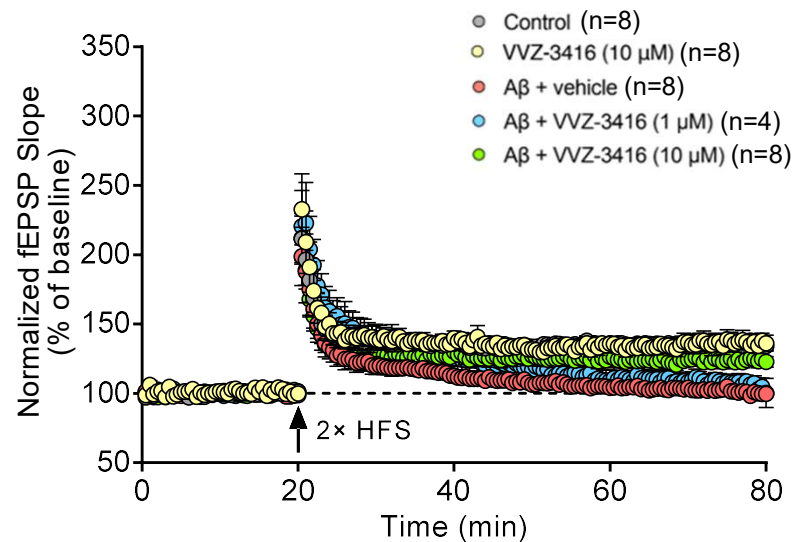
❖ Promotion of neurite outgrowth (PC12, 120 h)



• NGF: nerve growth factor

VVZ-3416 supports both neuronal survival and neurite extension in cellular models

❖ Restoration of A β -impaired LTP



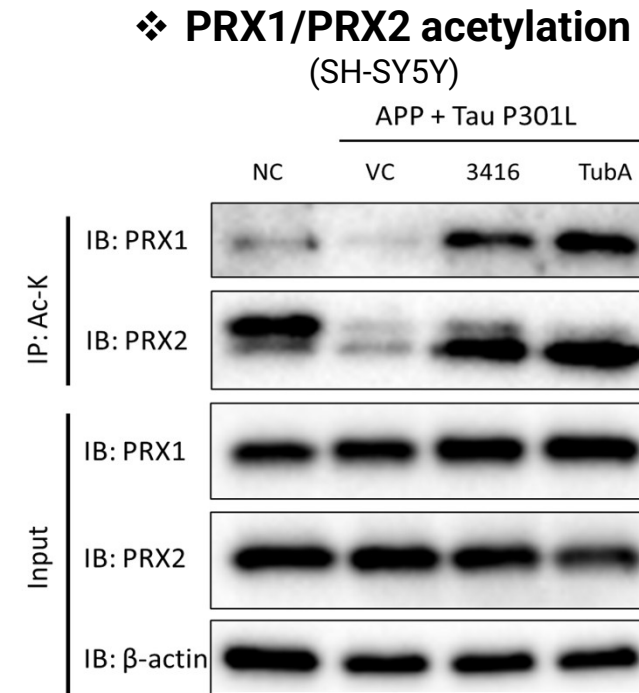
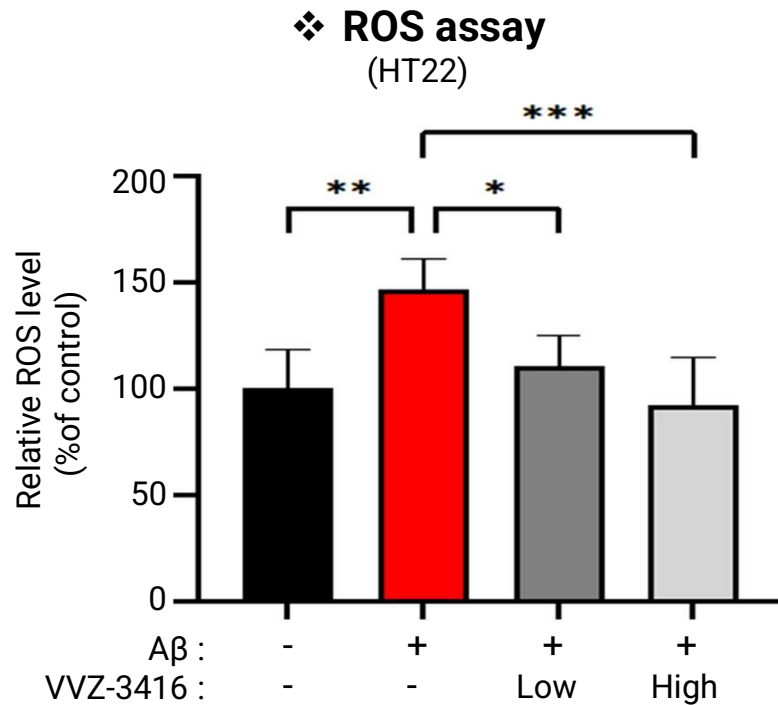
* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$, **** $p < 0.0001$

Improved LTP demonstrates functional preservation of synaptic networks beyond neuronal survival and neurite extension.

VVZ-3416 Mitigates Oxidative Stress Through Redox Regulation

HDAC6

MAO-B



HT22: Mouse hippocampal neuronal cell line, APP: Amyloid Precursor Protein; Tau P301L: pathogenic tau mutant; ROS: Reactive Oxygen Species; PRX1/PRX2: Peroxiredoxin 1/2 antioxidant enzymes; Ac-K : Acetyl-Lysine;TubA: Tubastatin A (selective HDAC6 inhibitor).

*p<0.05, **p<0.01, ***p<0.001, n=5-6/group

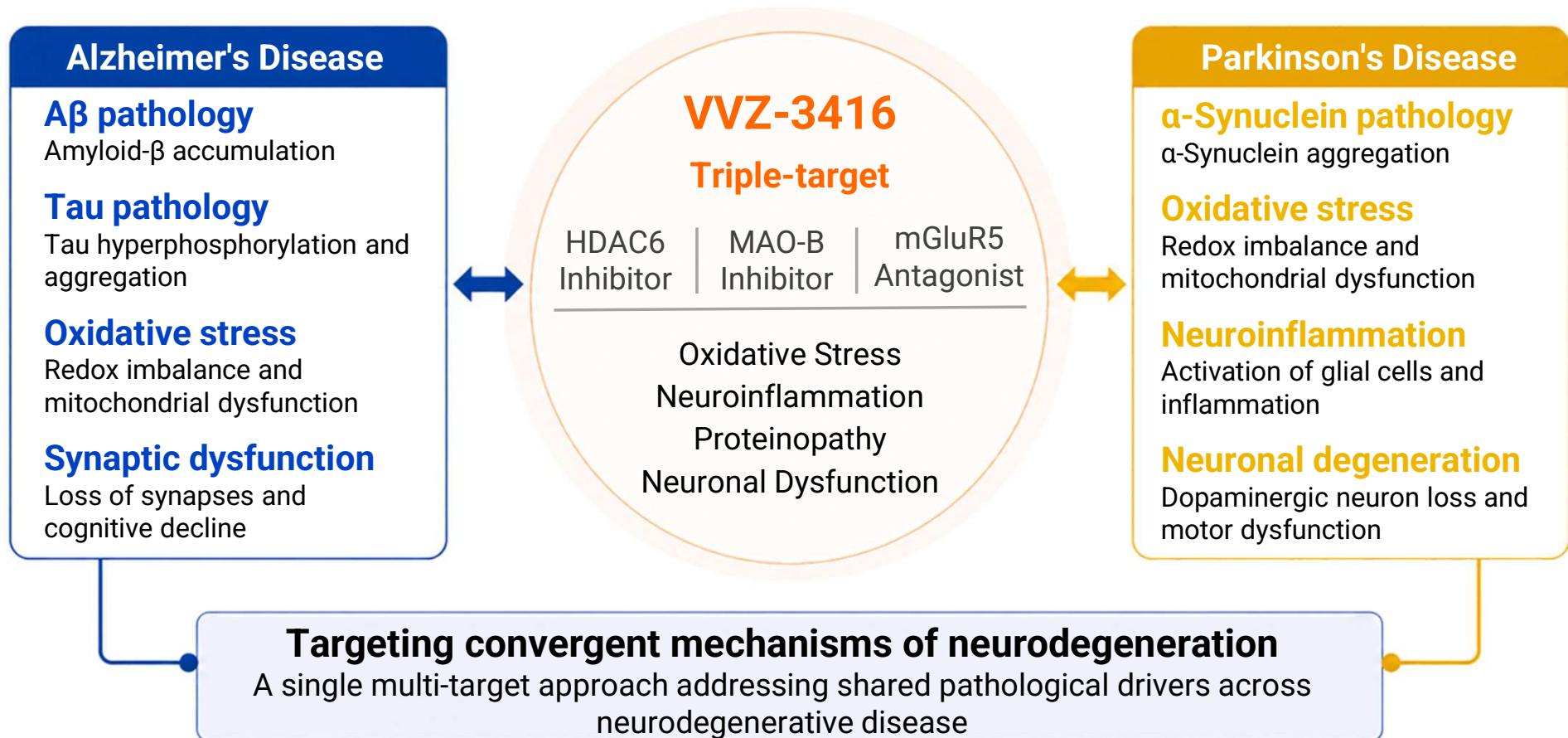
VVZ-3416 reduces oxidative stress through restoration of cellular redox homeostasis

SECTION

03

Preclinical Efficacy : Parkinson's Disease

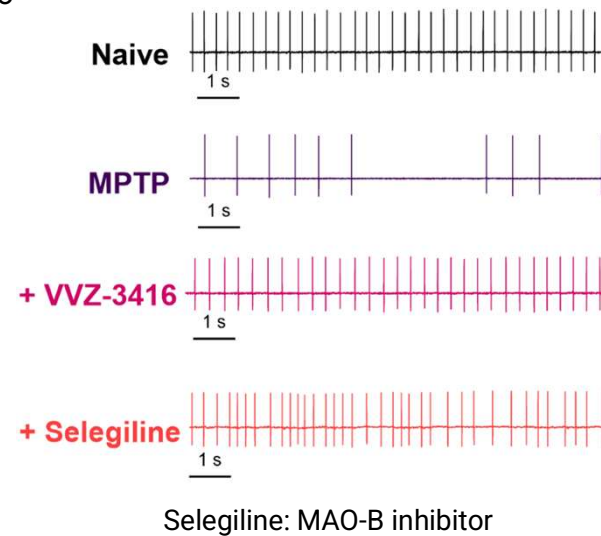
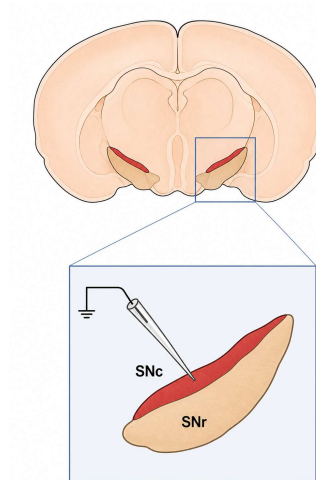
Shared Pathological Drivers Across Neurodegenerative Diseases



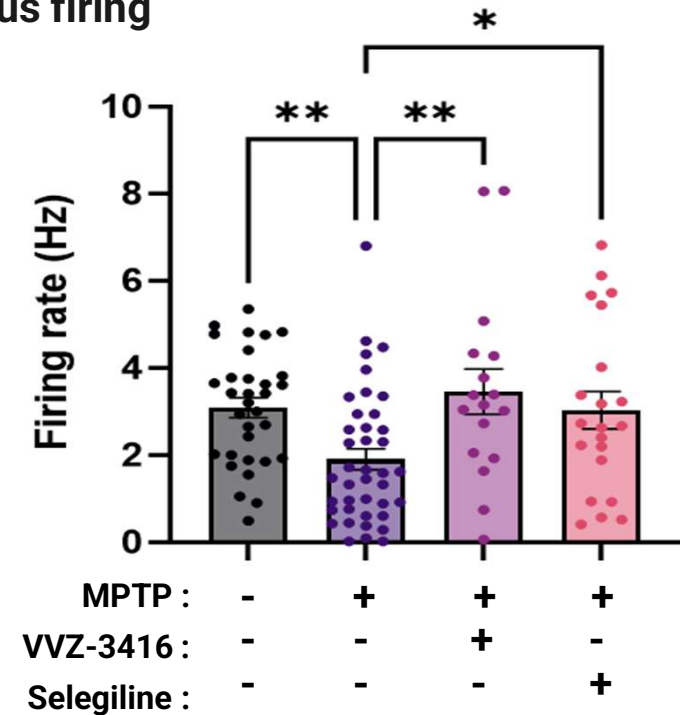
Effects of VVZ-3416 on SNc neuronal excitability in MPTP PD model

❖ Extracellular recording (cell-attached configuration): Spontaneous firing

Ex vivo recording of dopaminergic in SNc



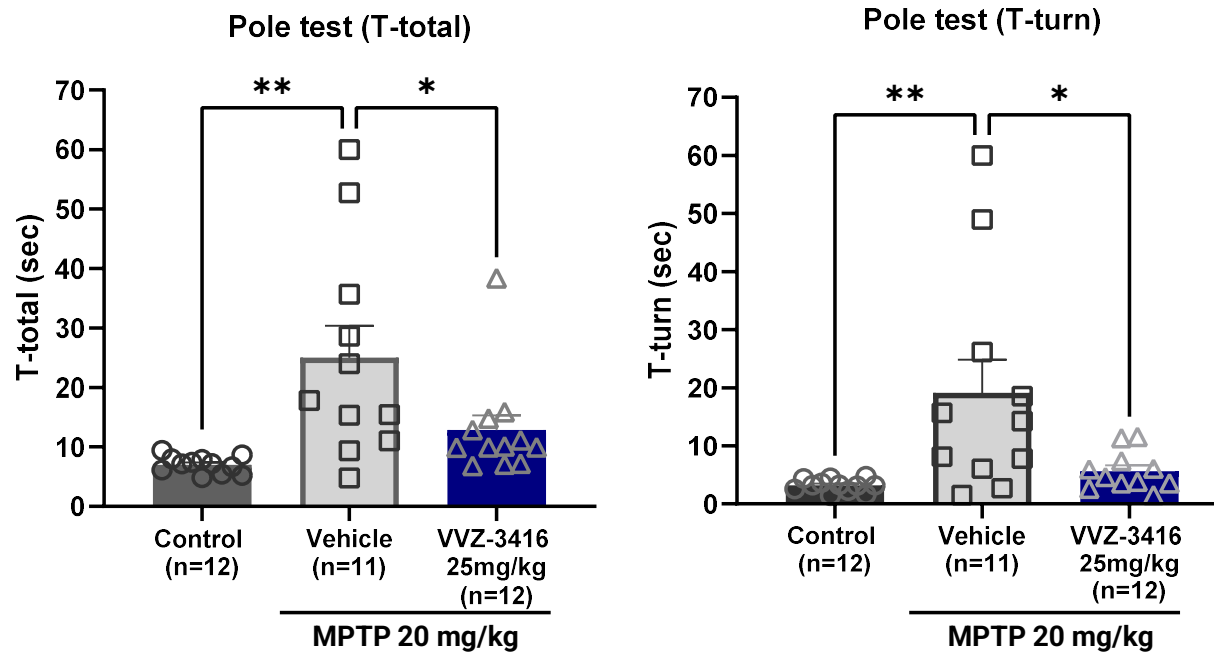
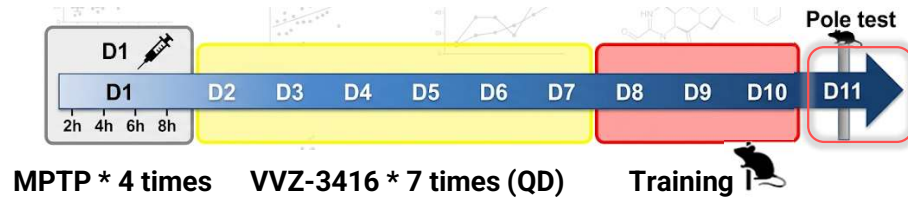
Selegiline: MAO-B inhibitor



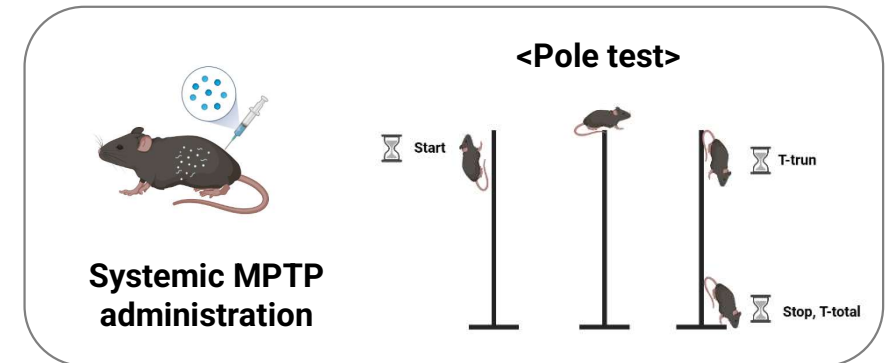
N (animals) / n (cells): naive N=3, n=32 · MPTP N=13, n=40 · +VVZ-3416 N=6, n=17 · +Selegiline N=7, n=21
Pre-incubation: VVZ-3416 10 μ M, Selegiline 0.1 μ M ($\geq$ 1 hr) | *p<0.05, **p<0.01

VVZ-3416 increases spontaneous firing rate impaired by MPTP

In Vivo Efficacy : MPTP-induced Mouse PD Model



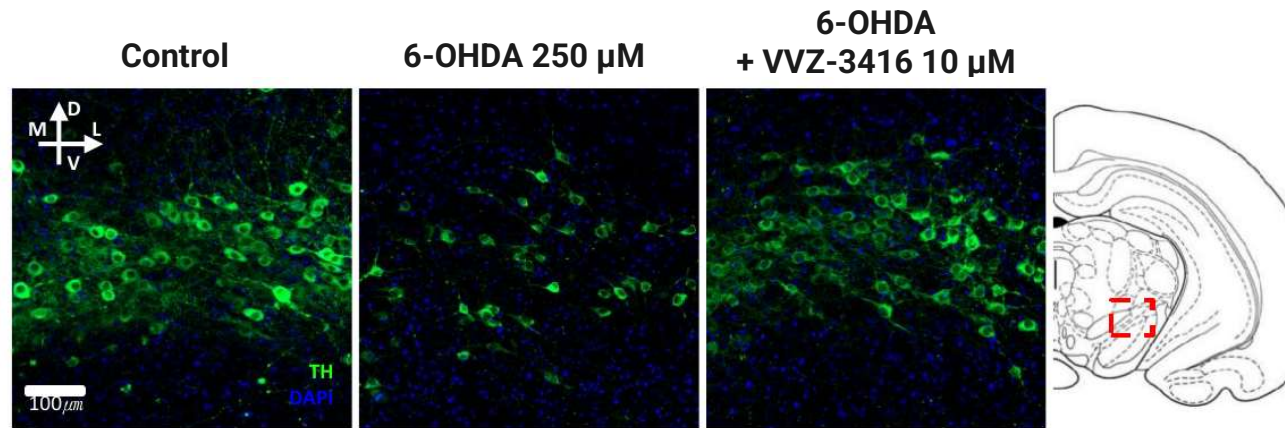
*p<0.05, **p<0.01



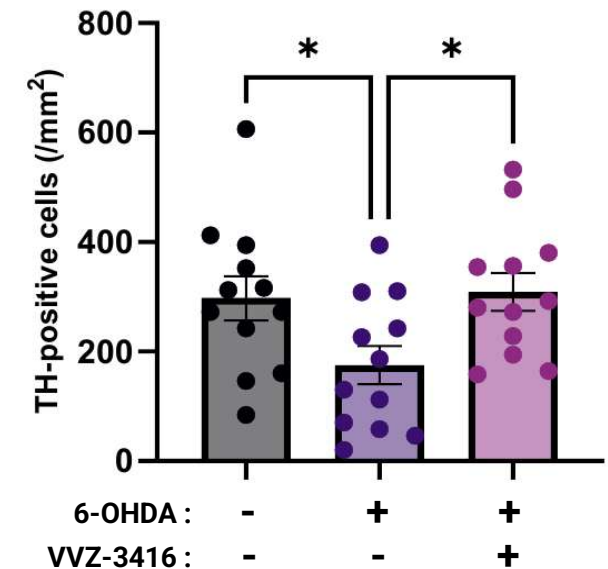
✓ **VVZ-3416 significantly reversed MPTP-induced motor deficits, as assessed by the pole test**

Effect of VVZ-3416 on SNpc TH⁺ Neuron Loss in Midbrain Slices

❖ Immunohistochemistry (TH staining) - 6-OHDA-induced dopaminergic degeneration model (OBSC)



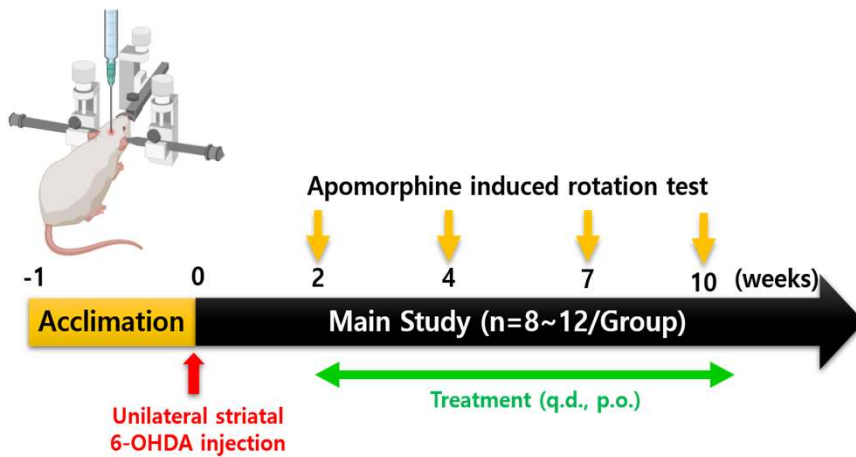
Treatment: 4 hr VVZ-3416 (10 μ M) pre-incubation → 24 hr co-treatment with 6-OHDA (250 μ M) + VVZ-3416 (10 μ M) at DIV11–12
N (independent slice cultures): Control N=12, · 6-OHDA N=12, · +VVZ-3416 N=12
*p<0.05



VVZ-3416 attenuates 6-OHDA-induced TH⁺ neuron loss in the SNpc region of organotypic midbrain slices.

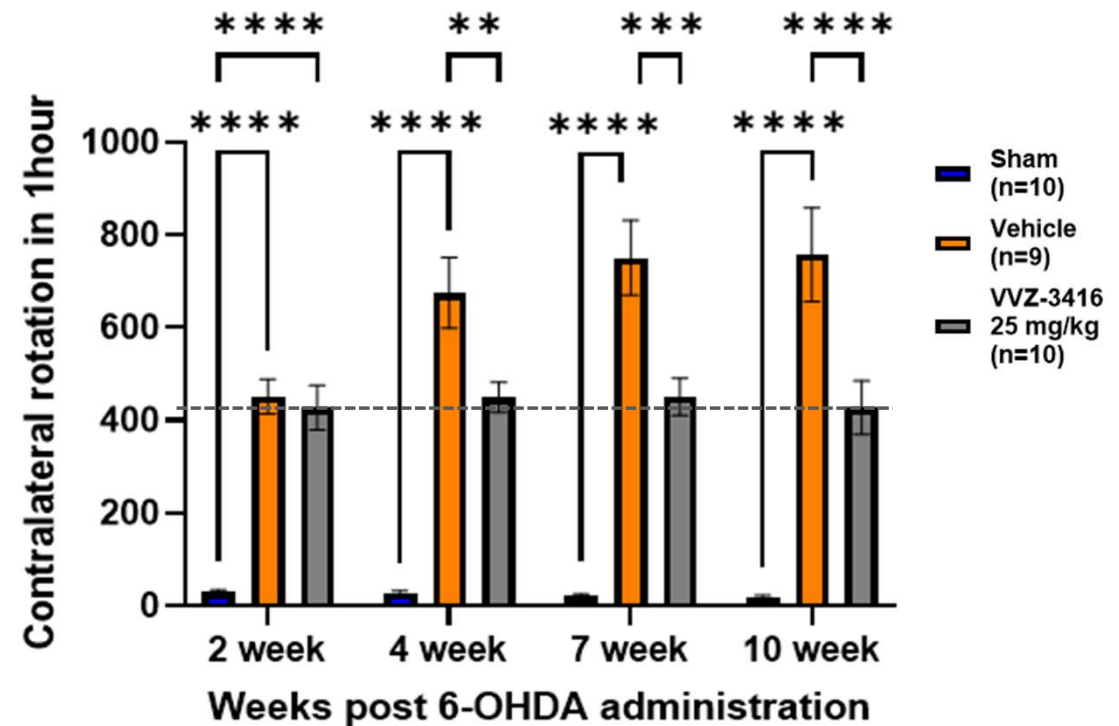
In Vivo Efficacy: 6-OHDA-induced Rat PD model

❖ 6-OHDA-induced rat PD model



“Prevention of disease progression”

❖ Apomorphine induced rotation test

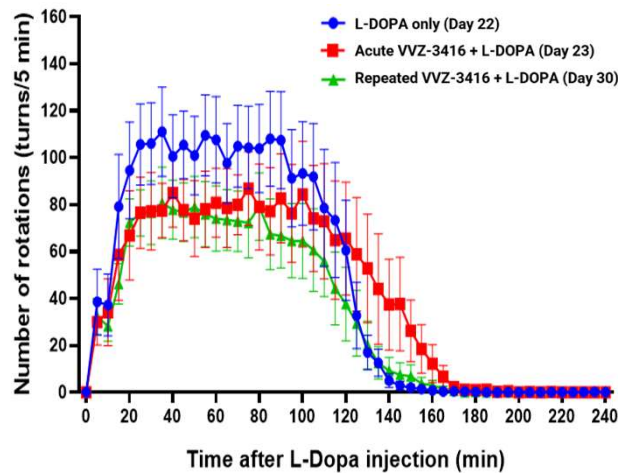


p<0.01, *p<0.001, ****p<0.0001 (n=8-12/group)

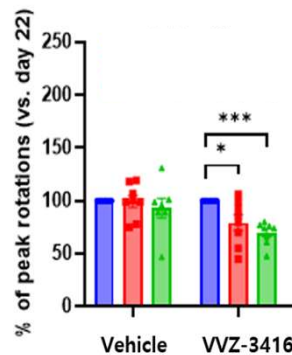
Exploratory Signal: PD-LID



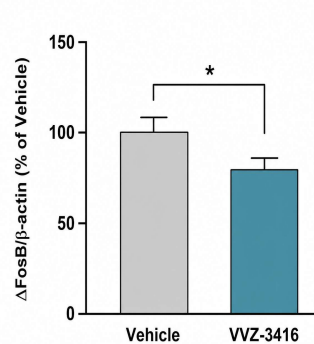
❖ 6-OHDA-Lesioned Rat Model of L-DOPA-Induced Dyskinesia



Peak rotations



Striatal Δ FosB



VVZ-3416 25 mg/kg p.o. administered as a single dose on Day 23 and following 7 days of repeated dosing on Day 30. Striatal tissue was subsequently collected for Δ FosB (n=8/group) | *p<0.05

Reduced peak rotational response and striatal Δ FosB while preserving antiparkinsonian efficacy

ON-TIME: No Change

Motor-effective window preserved (single & repeated)

PEAK ROTATION: ↓ Reduced

Significant decrease vs. baseline (p<0.05, both regimens)

STRIATAL Δ FosB: ↓ Reduced

Chronic LID biomarker; significantly lower vs. vehicle (p<0.05)

Current Limitation: AIMS score not yet assessed; reduced peak response has not been dissociated from motor sensitivity change.

Overview | Efficacy & MoA in Alzheimer's disease



Summary across target engagement, neuroprotection, anti-oxidation, tau clearance, synaptic plasticity, and in vivo efficacy studies

Study Category	Species / Model	Design / Readout	Key Results	Conclusions
In Vivo Target Engagement	3xTg-AD Mouse	Oral QD (25mg/kg) 7-week treatment	• Hippocampal Ac- α -Tubulin $\uparrow$ (WB)	Brain HDAC6 engagement
Neuroprotection (In Vitro & Ex Vivo)	SH-SY5Y OHSC (Hippocampal slice)	α A β -induced neurotoxicity	• Cell viability restored; LDH release $\downarrow$ • CA1 PI intensity & LDH $\downarrow$ (ex vivo)	α A β neuroprotection
Anti-Oxidant MoA	HT22 (α A β)	DCF-DA ROS assay	• Cellular ROS $\downarrow$	Redox protection
	SH-SY5Y (Tau mut)	Acetyl-IP assay	• Ac-Prx1, Ac-Prx2 $\uparrow$	
Tau Pathology Modulation	SH-SY5Y (α A β , Tau mut)	p-Tau WB Acetyl-IP assay	• p-Tau $\downarrow$ • HSP90 acetylation $\uparrow$	Tau degradation via HSP90 acetylation
Synaptic Function & Plasticity	PC12	Neurite outgrowth assay	• Neurite length & branching $\uparrow$	Restored synaptic plasticity
	Acute Brain Slice	α A β -impaired LTP	• Full LTP rescue (Basal unaffected)	
In Vivo Efficacy (Behavioral Rescue)	3xTg-AD Mouse	5-wk dosing → MWM (Wk 6), PAT (Wk 7)	• MWM learning latency $\downarrow$ • PAT retention latency $\uparrow$ (trend)	In vivo cognitive efficacy
In Vivo Pathology (Disease Modifying potential)	3xTg-AD Mouse	7-wk dosing, Hippocampal analysis	• CA1 p-Tau & A β $\downarrow$ (IHC) • Insoluble A β 1-42 & Tau $\downarrow$ (WB/ELISA)	Pathology reduction (in vivo)

Overview | Efficacy & MoA in Parkinson's disease



Summary across in vivo motor efficacy, L-DOPA/LID dynamics, DA neuroprotection, SNc firing, tonic GABA, and STN hyperexcitability studies

Study Category	System / Model	Design / Readout	Key Results	Conclusions
In Vivo Motor Function	Acute MPTP mouse model	Pole test Open-field test	• Pole test: T-turn & T-total ↓ • OFT: Total distance restored	Cross-model motor efficacy
	Unilateral 6-OHDA rat model	Apomorphine-induced rotation	• Apo rotation ↓	
L-DOPA-induced dyskinesia	Chronic L-DOPA-primed 6-OHDA rat model	Peak rotation counts Motor on-time duration Striatal ΔFosB	• L-DOPA peak rotation ↓ • On-time ↔ (NS) • STRIATAL ΔFosB ↓	Reduced L-DOPA sensitization
Neuroprotection	6-OHDA, Midbrain slice	SNpc TH loss	• SNpc TH(+) neuron loss attenuated	Dopaminergic neuroprotection
SNc Firing (MAO-B)	MPTP acute brain slice	Spontaneous firing	• SNc DA firing rate restored toward Naïve level	DA activity normalization
Tonic GABA (MAO-B)	Putrescine / MPTP-related acute slice paradigms	Tonic GABA current	• Astrocytic tonic GABA current normalized	Tonic inhibition normalization
STN Excitation (mGluR5)	Acute STN slices	DHPG-induced Ca ²⁺ transient and membrane depolarization	• Ca ²⁺ influx & depolarization ↓	STN mGluR5 antagonism

SECTION

04

Safety Pharmacology & Toxicology

Nonclinical Safety Package Overview



Summary across acute, repeat-dose, genotoxicity, and safety pharmacology studies (GLP)

Study Category	Species / Model	Study Design	Key Result	Conclusion
Single-Dose (Acute) Toxicity	Rat (po)	Rat 500 / 1000 / <u>2000 mg/kg</u>	ALD > 2000 mg/kg	Well tolerated
Repeat-Dose Toxicity (4-weeks)	Rat & Monkey (po)	Rat 50 / 100 / <u>200 mg/kg</u> Monkey 30 / 60 / <u>120 mg/kg</u>	Rat NOAEL 200 mg/kg; ≥17x margin Monkey NOAEL 120 mg/kg; ≥18x margin	Wide safety margin
Genotoxicity	AMES CA MN test	Bacterial Reverse Mutation Assay In vitro Chromosome Aberration Test In vivo bone marrow micronucleus assay	Negative across all three genotoxicity assays	Non-genotoxic
Safety Pharmacology (CNS / Respiratory)	Rat	Respiratory 50/100/200 mg/kg CNS FOB 50/100/200 mg/kg	No treatment-related effects observed	Clean profile
Safety Pharmacology (Cardiovascular)	in vitro hERG Dog telemetry Monkey telemetry	hERG Dog 10/20/40 mg/kg Monkey 30/60/120 mg/kg	IC ₅₀ 15 µM Dog ; no QT/hemodynamic liability Monkey ; Scheduled	Favorable

Acute and repeat-dose safety margins are wide, with no evidence of genotoxicity or CNS, respiratory, or cardiovascular liability

Target Organ & General Toxicity Findings



4-Week GLP Repeat-Dose Toxicity Study – Rat & Monkey

Species	Parameter	Finding	Onset (Dose Level)	Status
Rat	Target organ	Thymus ; Minimal to mild cellularity decrease of lymphocyte in cortex	All Dose Levels	Reversible
Rat	General toxicity	Activity ↓, Body weight and food consumption ↓, NEUT ↓	High dose (females)	Reversible
Monkey	Target organ	Thymus weight ↓, Organ-body and organ-brain ratios ↓ Mild to moderate in cortical and medullary lymphocyte cellularity ↓	All Dose Levels	Reversible
Monkey	General toxicity	No treatment-related changes	All Dose Levels	N/A

2/2

Species with NOAEL
Established Rat / Monkey

≥17x / ≥18x

Safety Margin
(Rat / Monkey)

0

Irreversible or
Severe Findings

All findings across dose levels, including at high doses, were mild and fully reversible



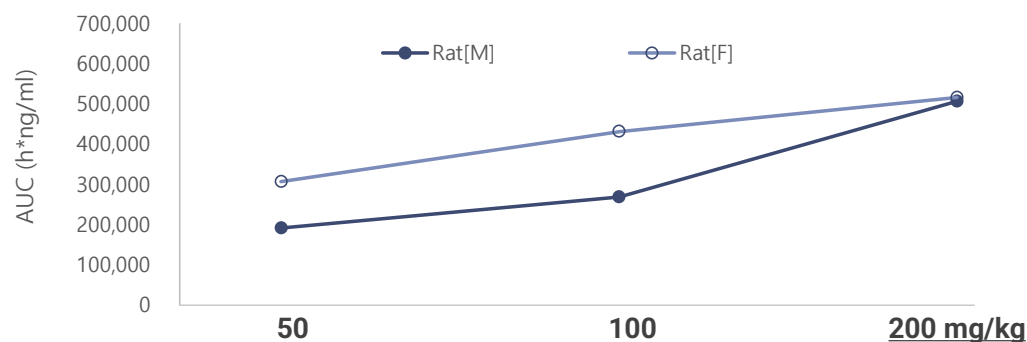
Toxicokinetics Support Dose-Proportional Exposure



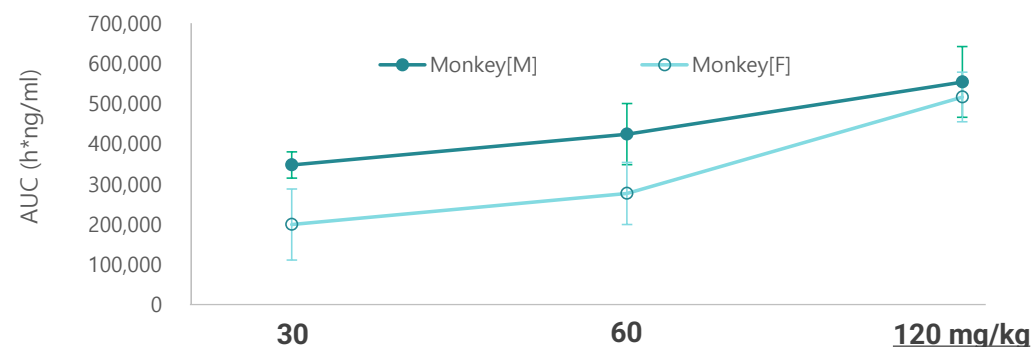
4-Week GLP repeat-dose study – Rat & Monkey systemic exposure (TK) summary

AUC Exposure vs. Dose

Rat TK AUC



Monkey TK AUC



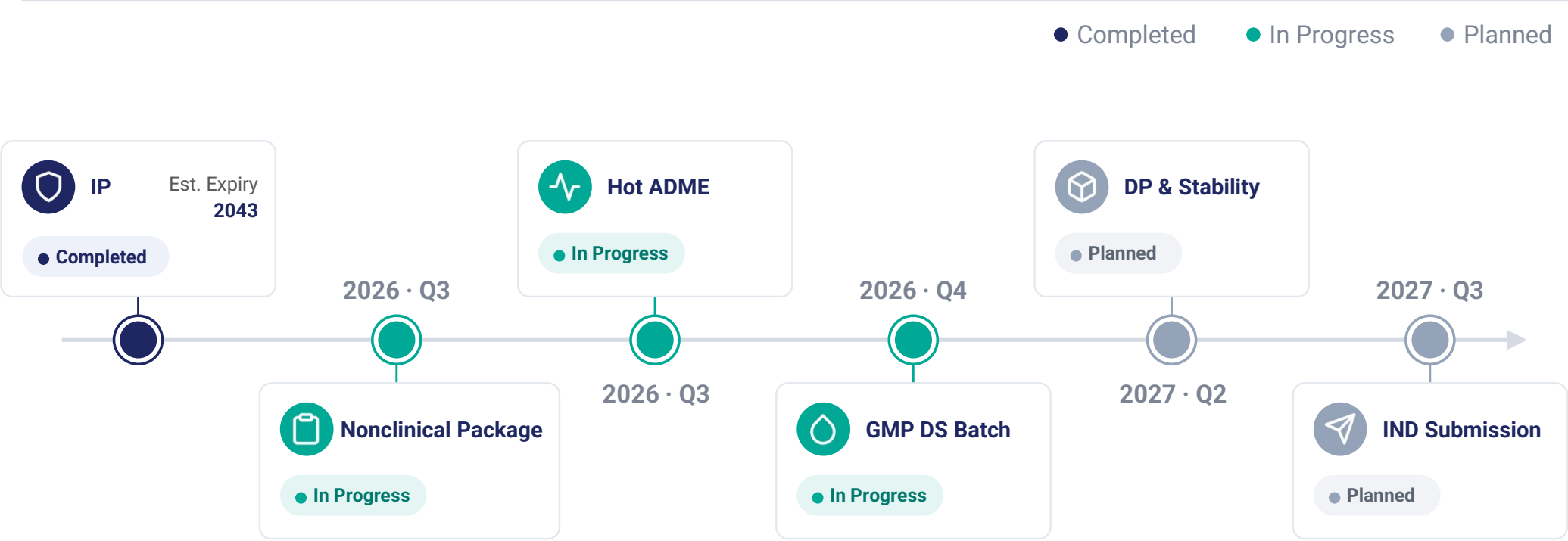
Parameter (NOAEL)	Rat – Male	Rat – Female	Monkey – Male	Monkey – Female
T _{max} (h)	4.0	8.0	2.8	3.6
C _{max} (ng/mL)	17,252	25,087	32,189	32,366
AUC _{0-24h} (ng·h/mL)	505,912	515,275	553,884	516,711
Accumulation Ratio	≈ 2.3 x	≈ 2.1 x	≈ 1.3 x	≈ 1.9 x
Gender Difference (M/F)	[Low ~ High Dose] 0.6~1.0 x		[Low ~ High Dose] 1.1~1.7 x	

Exposure increased **dose-proportionally with no gender difference**, with accumulation observed upon repeated dosing

Timeline for Clinical trial



PATH TO IND



Thank you

Partnering Contact:

Suk Hong

Head of Business Development

suk@vivozon.com



SECTION

05

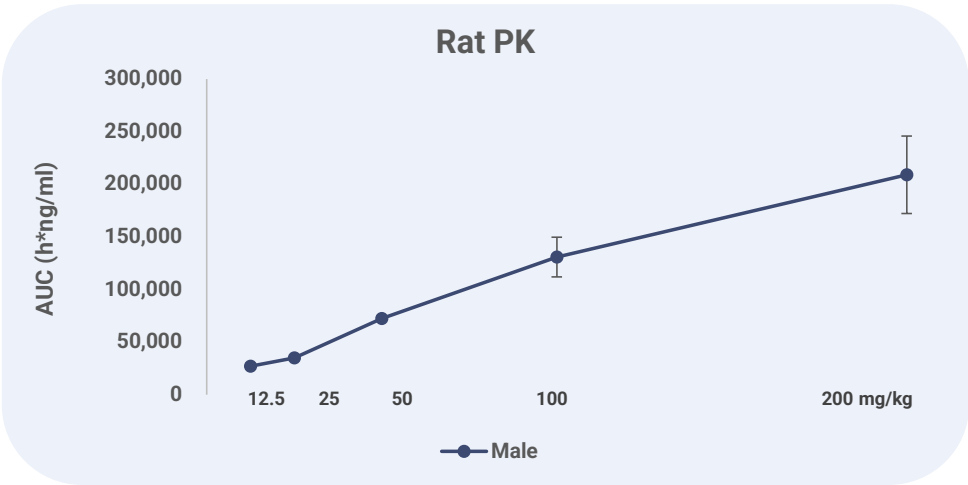
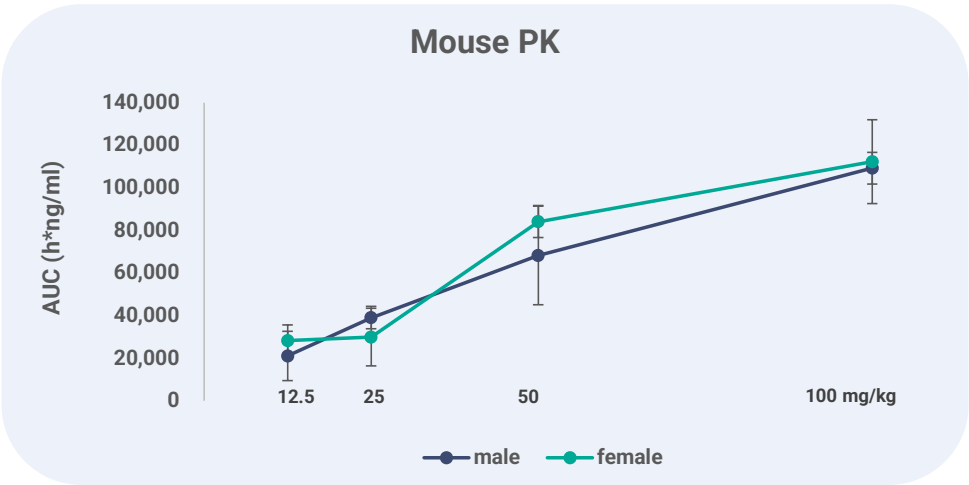
Appendix

Favorable PK Profiles in mice and Rats



Mouse & Rat systemic exposure (PK) summary

AUC Exposure vs. Dose



Mouse Male PK	12.5	25	50	100
T _{max} (h)	2.2	4.0	4.7	6.7
C _{max} (ng/ml)	1,671	2,733	4,529	7,530
AUC _{last} (h*ng/ml)	21,105	39,084	68,334	109,245

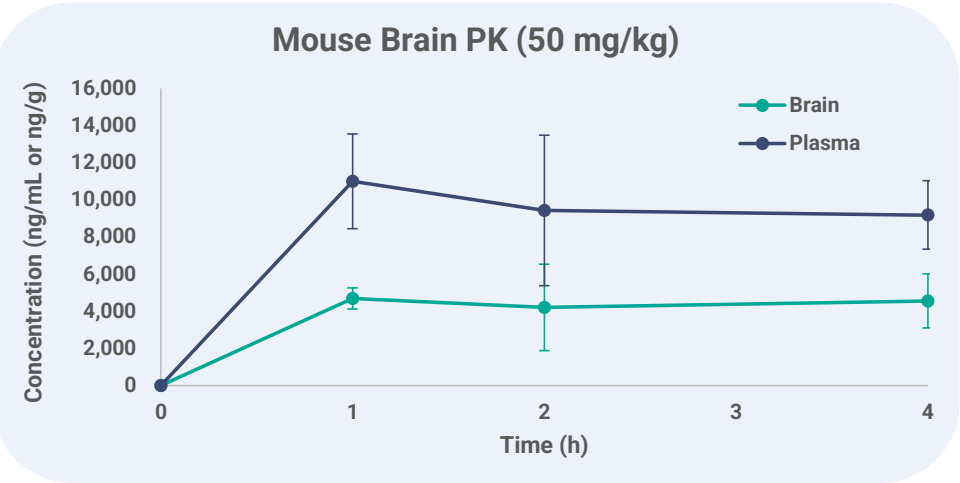
Rat Male PK	12.5	25	50	100	200
T _{max} (h)	4.0	2.7	5.3	12	8
C _{max} (ng/ml)	1,762	2,107	3,495	4,519	7,220
AUC _{last} (h*ng/ml)	27,195	35,064	72,503	130,938	209,042

Dose-proportional exposure observed across mouse and rat PK studies, supporting predictable pharmacokinetics and dose selection

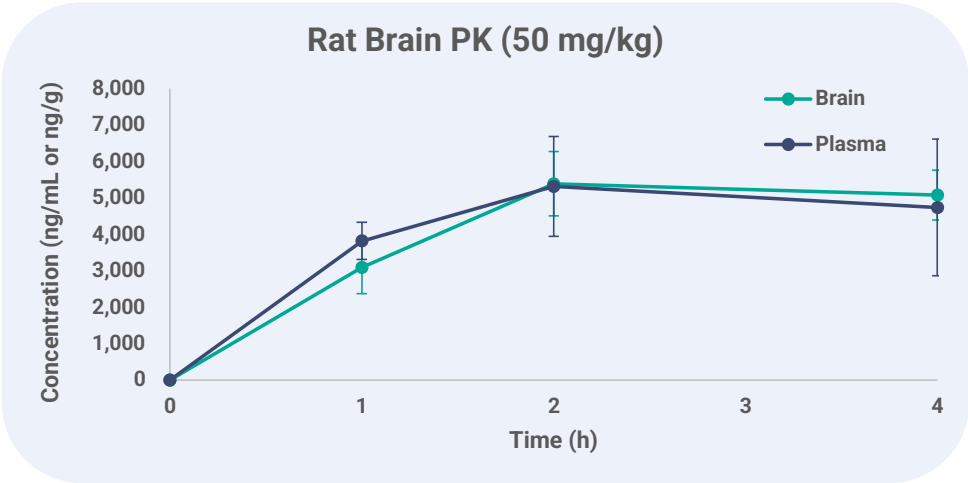
Brain Exposure in Mice and Rats



Mouse & Rat Brain PK Summary



Mouse Male	1	2	4
B/P ratio	0.43	0.44	0.49
Brain (ng/g)	4,695	4,208	4,561
Plasma (ng/ml)	10,997	9,438	9,183



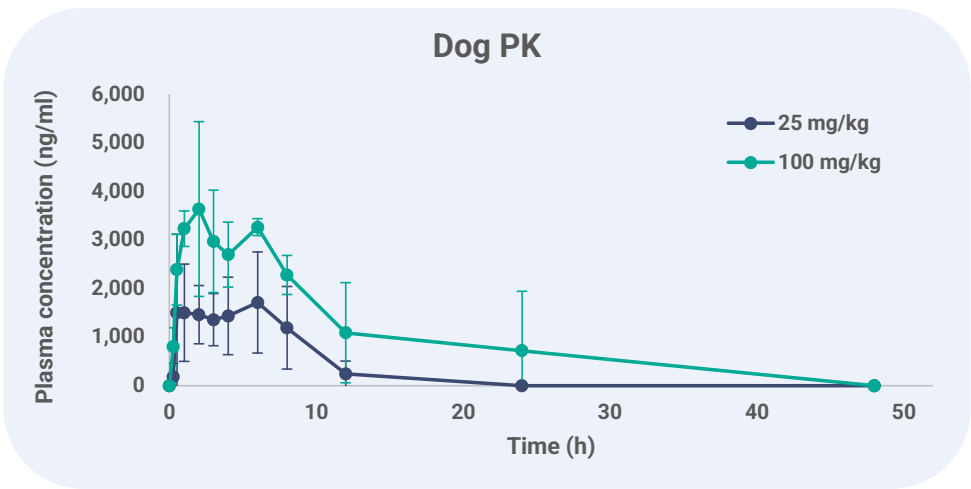
Rat Male	1	2	4
B/P ratio	0.82	1.03	1.14
Brain (ng/g)	3,096	5,386	5,077
Plasma (ng/ml)	3,819	5,314	4,737

Robust brain exposure across mouse and rat studies, supporting a favorable CNS profile and potential for effective target engagement

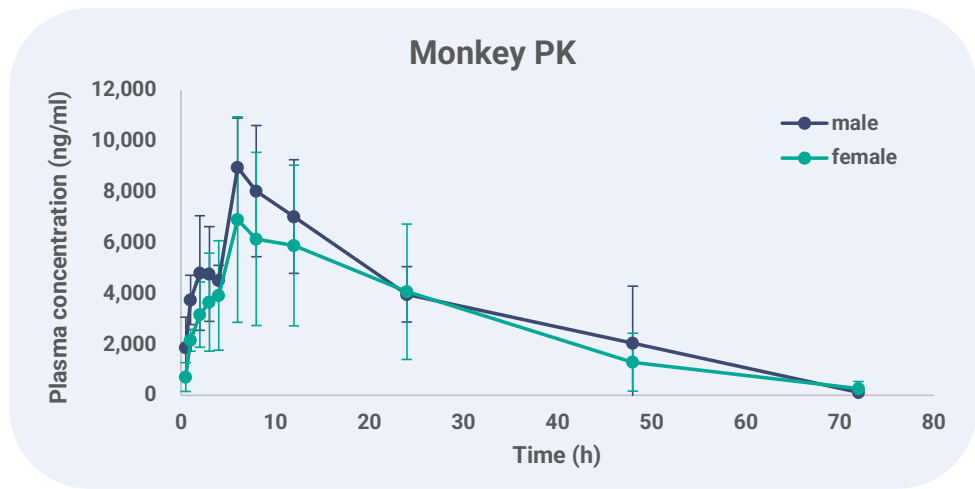
Favorable PK Profiles in Dogs and Monkeys



Dog & Monkey systemic exposure (PK) summary



Dog Male PK	25 mg/kg	100 mg/kg
T _{max} (h)	2.5	1.3
C _{max} (ng/ml)	2,051	4,196
AUC _{last} (h*ng/ml)	15,448	62,296



Monkey PK	Male 50 mg/kg	Female 50 mg/kg
T _{max} (h)	6	8
C _{max} (ng/ml)	8,960	7,188
AUC _{last} (h*ng/ml)	235,686	200,357

Dose proportionality was confirmed in dogs, with favorable exposure and no sex differences in monkeys

ADME Profile Supports a Drug-Like Candidate



In vitro absorption, distribution, metabolism characterization

Domain	Parameter	Result	Status
Absorption	Bioavailability (%F)	M 60% / R 79% / D 87%	Favorable
	Cell Perm. & P-gp Substrate	High Permeable, no efflux issue	High permeability
Distribution	Brain to Plasma ratio	M 0.4~0.5 / R 0.8~1.1	Good BBB Penetration
	Plasma protein binding (% Bound)	H 99 / M 88 / R 99 / D 99 / P 99	Favorable
	Brain tissue binding (% Bound)	H 98 / R 98 / D 98	Favorable
Metabolism	Microsomal stability (% Remaining)	H 87 / M 87 / R 89 / D 83 / P 99	Metabolically stable
Drug-Drug Interaction	CYP Inhibition (9 Major CYP Isoforms)	IC50 > 20 μ M (all)	Low DDI risk
	CYP Reaction Phenotyping	Major metabolism mediated by CYP1A1 (7% remaining at 60 min)	Defined CYP Pathway

The compound demonstrates a favorable drug-like ADME profile, characterized by good permeability, metabolic stability, and low CYP-mediated DDI risk.